AI Bootcamp Week 3: Model Invocation, Adversarial Agents, and Bounded Agents

Friday — Student Exercise Guide

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OBJECTIVES

By end of Week 3, the candidate can:

- Explain the DLP automation spectrum and what evidence is required to move up each step
- Explain why the DLP defaults to skills with `disable-model-invocation: true`, and articulate the three graduation criteria for enabling model invocation
- Write invocation test cases (positive trigger, negative trigger, quality) and use them to make a documented enable-or-gap decision on a real skill
- Build an adversarial agent that challenges a skill's output, and explain how it complements (not duplicates) a Python evaluation script
- Run an evaluator-vs-adversarial comparison and use the delta to improve either the EC or the skill
- Compose skills into a bounded agent: a multi-step pipeline run in one IDE session with evaluation and adversarial review at key steps
- Explain context window degradation, why bounded agents hit a ceiling, and what a custom harness is (awareness level; building one is Stage 4 work)
- Describe the IIOS stage framework: where DLP Completion sits (Stage 3, Task) and what Stage 4 (Workflow) requires

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TIMING

- **Monday:** 1:00 PM – 5:00 PM (travel morning: 4 hours)
- **Tuesday–Thursday:** 9:00 AM – 5:00 PM with 1-hour lunch (7 hours each)
- **Friday:** 9:00 AM – 12:00 PM (travel afternoon: 3 hours)

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FRIDAY (9:00 AM – 12:00 PM): SYNTHESIS AND ARTIFACTS REVIEW

9:00 – 10:30 AM: THE SPECTRUM REVISITED

Where does each of your Week 1 and Week 2 artifacts sit on the automation spectrum now? For each skill you built:

- Has it earned model invocation (three graduation criteria + EDD prerequisite + invocation test cases), or is there a documented gap?
- Does it have an adversarial agent, and has the evaluator-vs-adversarial delta been examined?
- If placed in a bounded agent chain, which steps upstream and downstream does it depend on, and where would you put the evaluation checkpoints?

Answer these questions in writing. This document is part of your certification gate artifacts.

Then, the coaching deliverable: write a 1-paragraph explanation, for a technically-aware manager, of why each step up the automation spectrum requires evaluation evidence before the next one is safe. Why you would not enable model invocation on an unmeasured skill, and why you would not chain skills into a bounded agent before each one passes its own evaluation. No jargon. This becomes part of your coaching vocabulary, and you will deliver it out loud.

10:30 – 11:30 AM: WEEK 3 ARTIFACTS REVIEW

- [ ] Graduation criteria document for at least one skill (all three criteria answered in writing)
- [ ] Invocation test cases (3 verified positive trigger + 2 quality, plus negative cases) for at least one skill
- [ ] Either a skill with model invocation enabled and test cases committed, or a gap specification document (both valid; vague "not ready" is not)
- [ ] Adversarial prompt for one Week 2 skill, run against 2–3 real outputs, findings documented and classified
- [ ] Evaluator-vs-adversarial comparison document with the delta noted
- [ ] Bounded agent run: full pipeline session output with evaluation and adversarial review at key steps
- [ ] Automation spectrum assessment: written evaluation of Week 1–2 artifacts against the spectrum criteria
- [ ] Manager-audience explanation paragraph: why each spectrum step requires evaluation evidence

11:30 AM – 12:00 PM: PREPARATION FOR WEEK 4

Brief preview of Week 4: the certification gate, the player-coach model, and what the resistance scenarios look like in practice. Candidates should arrive at Week 4 with all artifacts from Weeks 1–3 reviewed and committed.

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WEEK 3 DELIVERABLES (COMMITTED TO REPO)

- [ ] Graduation criteria document + invocation test cases for at least one skill
- [ ] Skill with model invocation enabled, or gap specification document
- [ ] Adversarial agent prompt with documented findings (classified: EC gap, skill gap, or false positive)
- [ ] Evaluator-vs-adversarial comparison document
- [ ] Bounded agent run artifact (full pipeline, evaluation and adversarial review at key steps, context window degradation observations noted)
- [ ] Automation spectrum assessment document
- [ ] Manager-audience explanation paragraph

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